

DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING

Program: B. Tech. (Electronics and Computer Engineering)							Semester: II		
Course: IoT Innovation with Arduino							Code:ECVS202		
Teaching Scheme (Hrs/week)				Evaluation Scheme (Marks)					
Lecture	Practical	Tutorial	Credit	CIE	ETE	TW	OR	PR	Total
-	04	-	02	-	-	25	-	-	25
Prerequisites:									
C/C++ and Python, familiarity with hardware components such as sensors and microcontrollers, and knowledge of data analytics and cloud computing concepts.									
Course Objectives:									
1. To provide a comprehensive introduction to IoT covering fundamentals, architecture, platforms, real-world examples, components, communication technologies, and challenges. 2. To develop proficiency in simulating Arduino Uno projects, including understanding its architecture, utilizing the IDE for Arduino software development, implementing Arduino libraries 3. To familiarize participants with sensor and actuator operations through hands-on experience with Arduino, including interfacing various sensors and actuators. 4. To familiarize participants with the fundamentals of wireless networking using the ESP8266 WiFi module, including understanding different Wi-Fi libraries, configuring a web server, and transmitting sensor data. 5. To compare and contrast M2M (Machine-to-Machine) communication with IoT (Internet of Things) and analyze the performance and suitability of MQTT, HTTP, and CoAP communication protocols. 6. To explore and understand the integration of virtualization concepts, cloud architecture, cloud computing benefits, and various cloud services, alongside IoT-specific platforms and protocols like Thing Speak API and MQTT, with a focus on interfacing ESP8266 microcontrollers with web services.									
Course Outcomes:									
CO1	Gain a foundational understanding of IoT concepts, enabling them to identify key components, discuss architecture and protocols, recognize various platforms and real-world applications, and understand challenges associated with IoT implementation.								
CO2	Gain practical skills in simulating and programming Arduino Uno projects, enabling them to create and test embedded systems before physical implementation virtually.								
CO3	Gain practical skills in integrating temperature, humidity, motion, light, gas sensors, relay switches, and servo motors with Arduino for real-world applications.								
CO4	Gain practical skills in setting up and managing basic wireless networks with ESP8266, deploying web servers, and integrating sensor data for web-based applications.								
CO5	Determine the optimal communication protocol for various IoT applications based on efficiency, scalability, and interoperability considerations.								
CO6	Gain practical knowledge and skills in designing and implementing cloud-based IoT solutions using virtualization, cloud services (SaaS, PaaS, IaaS), and IoT-specific platforms, enhancing proficiency in ESP8266 integration with web services for real-world applications.								

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Course Contents:		
Unit	Description	Duration (Hrs.)
1.	Introduction to IOT: Fundamentals of IoT, IoT Architecture and Protocols, Platforms for IoT, Real-world Examples of IoT, Overview of IoT, Components and Communication Technologies, Challenges in IoT	9
2.	Arduino Simulation Environment: Arduino Uno Architecture, Setting up the IDE and Writing Arduino Software, Arduino Libraries, Basics of Embedded C Programming for Arduino, Interfacing LED, Push Button, and Buzzer with Arduino, Interfacing Arduino with LCD.	9
3.	Sensor & Actuators with Arduino: Overview of Sensor Operation, Analog and Digital Sensors, Interfacing Temperature, Humidity, Motion, Light, and Gas Sensors with Arduino, Interfacing Actuators with Arduino, Interfacing Relay Switches and Servo Motors with Arduino.	10
4.	Basic Networking with ESP8266 Wi-Fi module: Fundamentals of Wireless Networking, Overview of the ESP8266 Wi-Fi Module, Different Wi-Fi Libraries, Web Server: Introduction, Installation, and Configuration, Sending Sensor Data to a Web Server.	10
5.	IoT Protocols: M2M (Machine-to-Machine) vs. IoT (Internet of Things), Communication Protocols MQTT, HTTP, and CoAP	9
6.	Cloud Platforms for IOT: Virtualization Concepts and Cloud Architecture, Benefits of Cloud Computing, Cloud Services: SaaS, PaaS, IaaS, Cloud Providers and Their Offerings, Overview of IoT Cloud Platforms, Thing Speak API and MQTT, Interfacing ESP8266 with Web Services.	9
TOTAL		56
List of Experiments		
- Exploring the Fundamentals and Architecture of IoT. - Real-World IoT Applications and Communication Technologies. - Interfacing and Programming Basic Components with Arduino. - Interfacing Arduino with an LCD Display. - Interfacing Sensors with Arduino. - Interfacing Actuators and Control Mechanisms with Arduino. - Evaluating Wi-Fi Module Performance. - Implementing Sensor Data Transmission via Web Server. - Comparative Analysis of Data Transfer Efficiency in M2M and IoT Networks. - Security Analysis of Communication Protocols in IoT Environments. - Evaluating Cloud Service Models. - Integration of IoT Devices with Cloud Platforms.		
Project Here are some guidelines to consider when planning and executing IoT projects: - Define Clear Objectives - Select Appropriate Technology		

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- 3) Design for Scalability
- 4) Ensure Data Security
- 5) Consider Interoperability
- 6) Focus on User Experience
- 7) Optimize Power Efficiency
- 8) Comply with Regulations
- 9) Test and Validate
- 10) Plan for Maintenance and Updates
- 11) Document and Share Knowledge

Engage Stakeholders

Text Books:

1. Kamal, R., "Internet of Things – Architecture and Design Principles," 1st Edition, McGraw Hill, 2017.
2. Simone Cirani, "Internet of Things - Architectures, Protocols and Standards," Wiley, 2018.
3. Alessandro Bassi, "Enabling Things to Talk - Designing IoT Solutions with the IoT Architectural Reference Model," Springer, 2013.

Reference Books:

1. Alasdair Gilchrist, "Industry 4.0: The Industrial Internet of Things," A press, 2016.
2. Sudip Misra, Chandana Roy, Anadarup Mukherjee, "Introduction to Industrial Internet of Things and Industry 4.0," CRC Press, 2020.
3. Giacomo Veneri, Antonio Capasso, "Hands-on Industrial Internet of Things," Packt Press, 2018.

E-Resources:

1. <https://www.arduino.cc/>
2. <https://www.tinkercad.com/>
3. https://www.tutorialspoint.com/internet_of_things/index.htm
4. <https://thingspeak.com/>
5. <https://www.nabto.com/guide-iiot-protocols-standards/>
6. <https://thingsboard.io/>